

NeuroAnalyzer report: Signal Characterization

NeuroAnalyzer v0.3.0 · MATLAB 2026a · 2026-09-25T05:28:13+00:00 · window image: exportapp

Signal Characterization

Default response features (latency, FWHM, AUC, rise/decay time) and compare groups with statistics

Help

Single file Groups & statistics

1 Load data

Load .mat file

Try demo data

Data type

LDF segments (segmentedLDF, ...)

Approx. LDF duration

LDF segments: 8 series, 1 to 10 Hz, -5 to 20 s

2 Parameters

Stimulus onset (s)

0

Baseline start (s)

-5

Baseline end (s)

0

Direction

Auto

3 Features

Peak latency

Onset delay (50%)

FWHM

AUC (positive)

Export features

4 Export results

Export to CSV / MAT

8 series, 8 features(s) computed

Save session...

Open session...

Report (PDF)...

Methods text...

Selected series

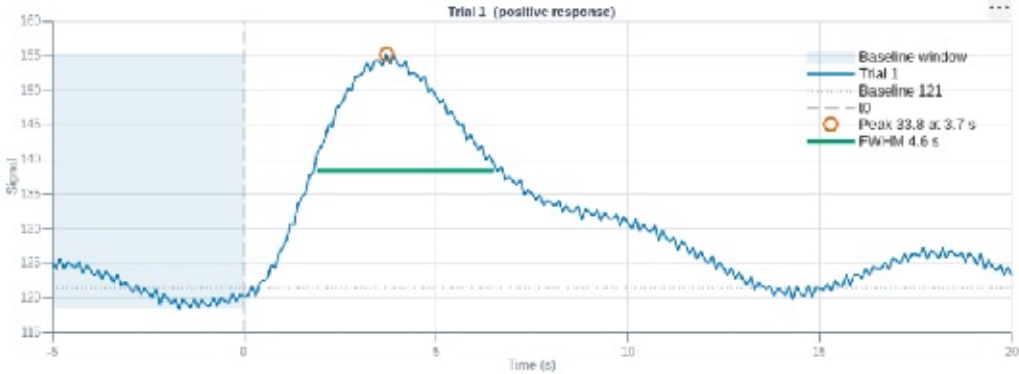
Show:

Trial 1

Clicked left: selected baseline window on plot for FWHM

PDF (vector)

Export figure...



Results (one row per series; NA = not computed or not found)

Trial	Channel	PeakLatency s	OnsetDelay s	FWHM s	AUCpos	AUCneg	RiseTime s	DecayTime s	PeakAmp	Integral
Trial 1		3.7000	1.5000	4.0000	219.7377	5.0527	2	2.9000	33.7043	2.5437e+03
Trial 2		3.8000	1.5000	4.0000	189.4461	13.7016	2.1000	2.7000	33.5007	2.6713e+03
Trial 3		4.2000	2.2000	4.0000	173.3666	15.0192	2.0000	2.5000	32.9732	2.6777e+03
Trial 4		4.2000	2.2000	4.0000	163.4176	23.1768	2.2000	2.8000	31.5663	2.6661e+03
Trial 5		4.2000	2.2000	4.0000	187.2848	27.6248	2.5000	2.8000	33.1207	2.6237e+03
Trial 6		4.2000	2.2000	4.0000	163.5077	27.3064	2.5000	2.8000	28.8085	2.5870e+03
Trial 7		4.2000	1.5000	3.1000	177.8855	13.0134	3.2000	3.4000	27.7585	2.5435e+03
Trial 8		3.7000	1.7000	6.5000	192.7184	11.4170	2.9000	4.5000	28.1098	2.5077e+03

Session saved to SignalCharacterizationApp.nasession.mat (L input file) with WDU, settings and results.

© Copyrights by Alejandro Suarez, Ph.D. · v0.3.0

Window: Signal Characterization (SignalCharacterizationApp)
NeuroAnalyzer v0.3.0
MATLAB 26.1.0.3346908 (R2026a) Update 5
OS: Linux 6.17.0-1022-azure (GLNXA64)
Session created: 2026-09-25T05:28:13+00:00
Report created: 2026-09-25T05:28:13+00:00

NOTES
SignalCharacterizationApp walkthrough: demo data

INPUTS (1)
1. Single file: demo_ldf_trials.mat
path: /tmp/NeuroAnalyzerDemo/demo_ldf_trials.mat
30,187 bytes, modified 2026-09-25T05:23:13
MD5 9c0c093692d06b01dc03e433e6695124

SETTINGS
mode = Single file
single.input = 1
single.dataType = LDF segments (segmentedLDF, segmentedTime)
single.t0 = 0
single.baseline = [-5 0]
single.direction = Auto
single.features = {Peak latency, Onset delay (50%), FWHM, AUC positive, AUC negative, Rise time, Decay time, Peak amplitude, Stim-response integral}
single.series = Trial 1
single.extracted = true
groups.order = {}
groups.groupName = Control
groups.feature = Peak amplitude
groups.subjectMode = File (mean trace)
groups.t0 = 0
groups.baseline = [0 0.05]
groups.baselineAuto = true
groups.direction = Auto
groups.design = Paired (same animals)
groups.method = Parametric
groups.groupA = -
groups.groupB = -
groups.plotStyle = Mean ± SEM
groups.tested = false

KEY RESULTS
Single file: 8 series (LDF segments)
PeakLatency_s: mean 4.225, SD 0.4367 (n = 8)
OnsetDelay_s: mean 2.025, SD 0.1982 (n = 8)
FWHM_s: mean 5.162, SD 0.7558 (n = 8)
AUCpos: mean 181.6, SD 21.23 (n = 8)
AUCneg: mean 15.96, SD 6.034 (n = 8)
RiseTime_s: mean 2.425, SD 0.44 (n = 8)
DecayTime_s: mean 3.062, SD 0.6718 (n = 8)
PeakAmp: mean 30.85, SD 2.492 (n = 8)
Integral: mean 2616, SD 63.09 (n = 8)